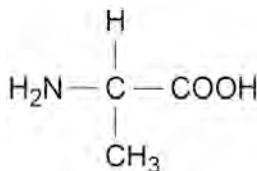


Answer **all** questions in the spaces provided.

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0 1 . 1 **Figure 1** shows the structure of the amino acid alanine.

Figure 1



Name the NH_2 and COOH groups in the amino acid.

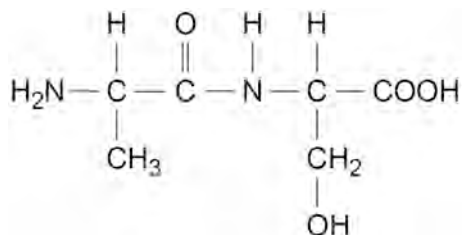
[1 mark]

NH_2 _____

COOH _____

Figure 2 shows the molecule formed from alanine and another amino acid.

Figure 2



0 1 . 2 Name the type of bond and the type of molecule formed when the two amino acids in **Figure 2** are joined.

[2 marks]

Type of bond _____

Type of molecule _____



0 1 . 3

Table 1 shows the R groups of six different amino acids.

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Table 1

Amino acid	R group	Amino acid	R group
Alanine	CH ₃	Glycine	H
Aspartic acid	CH ₂ COOH	Serine	CH ₂ OH
Glutamine	(CH ₂) ₂ CONH ₂	Valine	CH(CH ₃) ₂

Use the information in **Table 1** to identify the amino acid joined to alanine in **Figure 2**.

[1 mark]

Question 1 continues on the next page

Turn over ►

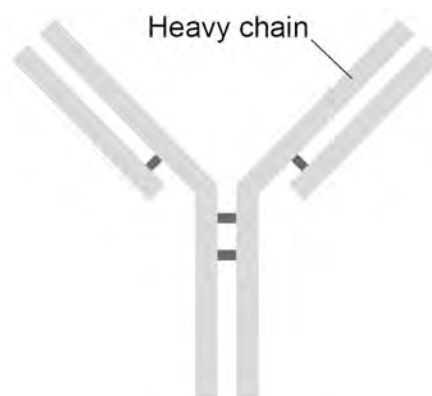


01.4

Figure 3 represents the structure of an antibody.

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Figure 3



The total number of amino acids in a particular antibody molecule is 1322

The ratio of amino acids in the heavy chains to the amino acids in the light chains is 2.13 : 1

Calculate how many amino acids there are in **one** heavy chain.

Give your answer to the nearest whole number.

[2 marks]

_____ amino acids

6

